Subhendu Bhandary

Postdoctoral Researcher
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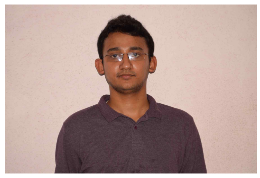

Curriculum Vitae

Personal Profile

Name **Subhendu Bhandary**
Date of birth **13 August, 1993**
Nationality **Indian**

Research Interests

I am interested in understanding the resilience of complex networks under the influence of environmental and demographic perturbations. I have worked on a variety of problems on complex networks ranging from ecological to neuronal networks. Sudden loss of network resilience is known to drive tipping points in natural systems, and hence increases the significance of resilience thinking. In fact, several anthropogenic factors are known to drive the collective dynamics of complex ecological systems which further hamper species persistence increasing the risk of community-wide extinction. Presently, I am working on how networks' structural properties interplay to alter systems' dynamics in static as well as time-varying networks. I have also been working to determine and mitigate tipping points in mutualistic networks in a degrading environment, such as investigating tippings in mutualistic plant-pollinator networks exposed to climate warming. In an ongoing project, I am working on preventing network extinction through strategic interactions via games among species sharing a common niche. I wish to explore further along this direction of research and other related ones with the aim of developing strategies/mitigation policies that can promote the resilience of ecological networks. The basic tools of my research are applied mathematics, dynamical systems, and numerical computations. I also have a working knowledge of theoretical ecology.

Education

2018-2023 **PhD in Applied Mathematics**, Department of Mathematics, IIT Ropar, Rupnagar, Punjab, India - 140 001
Supervised by **Dr. Partha Sharathi Dutta**

2015-2017 **Master of Science in Mathematics and Computing**, Indian Institute of Technology, Guwahati, Assam- 781039

2014-2015 **Master of Science in Applied Mathematics**, University of Calcutta, 87/1, College Street, Kolkata-700 073

2011-2014 **Bachelor of Science in Mathematics** , Ramakrishna Mission Residential College, Narendrapur, Kolkata, West Bengal 700103

RESEARCH EXPERIENCE

August 2025 – Continuing — **Postdoctoral Researcher**, Chair of Network Dynamics, Institute of Theoretical Physics and Center for Advancing Electronics Dresden (cfaed), TUD Dresden University of Technology, 01062 Dresden, Germany
Host: **Prof. Marc Timme**

June 2023 – May 2025 — **Postdoctoral Researcher**, Department of Evolutionary Biology and Environmental Studies, University of Zurich, Switzerland
Host: **Prof. Jordi Bascompte**

Fellowships and Awards

2017 **Cleared Graduate Aptitude Test in Engineering (GATE) with All India Rank 53**
Fellowship for pursuing PhD or M.Tech by the Ministry of Human Resource Development (MHRD), Government of India

2016 **Qualified NBHM M.Sc scholarship written exam.**

2015 **Cleared Joint Admission test for M.Sc (JAM) with All India Rank 198.**
Secured Admission in Indian Institute of Technology Guwahati.

2013 **Qualified MADHAVA Mathematics competition.**

Computer skills

Languages	C, C++, R, Python
Packages	MATLAB, Julia, MATCONT, XP-PAUT
Platforms	Linux, Windows

Languages

English	Fluent
Bengali	Native
Hindi	Fluent

Publications

Papers published, revised and in preprint

1. Narang, A., **Bhandary, S.,** Kaur, T., Gupta, A., Banerjee, T., and Dutta, P.S., *Long-range dispersal promotes species persistence in climate extremes*, **CHAOS**, AIP, Vol. 29, Issue 10, pp 103136 (2019). DOI: https://doi.org/10.1063/1.5120105
2. **Bhandary, S.**, Kaur, T., Banerjee, T., and Dutta, P.S., *Network resilience of FitzHugh-Nagumo neurons in the presence of non-equilibrium dynamics,* **Physical Review E**, APS, Vol. 103 (2), 022314 (12) (2021). DOI: https://doi.org/10.1103/PhysRevE.103.022314
3. **Bhandary, S.**, Biswas, D., Banerjee, T., and Dutta, P.S., *Effects of time-varying habitat connectivity on metacommunity persistence,* **Physical Review E**, APS, Vol. 106 (1), 014309 (12) (2022). DOI: https://doi.org/10.1103/PhysRevE.106.014309
4. Deb, S., **Bhandary, S.,** Sinha, S., Jolly, M.K., and Dutta, P.S., *Identifying Critical Transitions in Complex Diseases*, **Journal of Biosciences,** Springer, Vol. 47, 25, 1-16 (2022). DOI: https://doi.org/10.1007/s12038-022-00258-7

5. **Bhandary, S.**, Deb, S. and Dutta, P. S., *Rising Temperature Drives Tipping Points in Mutualistic Networks,* **Royal Society Open Science**, *The Royal Society Publishing, Vol. 10(2), 221363(17) (2023).* DOI: https://doi.org/10.1098/rsos.221363
6. Deb, S., **Bhandary, S.**, and Dutta, P. S., *Evading tipping points in socio-mutualistic networks via structure mediated optimal strategy,* **Journal of Theoretical Biology**, *Elsevier, Vol. 567, 111494(10) (2023).* DOI: https://doi.org/10.1016/j.jtbi.2023.111494
7. **Bhandary, S.**, Banerjee, T., and Dutta, P.S., *Nonautonomous perturbations and the stability of ecosystems,* **Physical Review E**, APS, Vol. 108(2):024301 (2023). DOI: https://doi.org/10.1103/PhysRevE.108.024301
8. **Bhandary, S.**, and Dutta, P.S., *Effects of adaptive niche based interactions due to strategic identities on metacommunity persistence.* **Proceedings of the Royal Society A**. DOI: https://doi.org/10.1098/rspa.2026.0009
9. **Bhandary, S.**, Gawecka, K. A., Pedraza,F., and Bascompte, J.,*Landscape configuration and community structure jointly determine the persistence of mutualists under habitat loss.* **Proceedings of the Royal Society B: Biological Sciences** DOI: https://doi.org/10.1098/rspb.2025.2602
10. **Bhandary, S.**, and Bascompte, J.,, *Local and Spatial Processes Shape the Collapse and Recovery of Mutualistic Networks* DOI: https://doi.org/10.1101/2025.11.25.690369 **(Under review)**
11. Deb, S., **Bhandary, S.**, Jolly, M.K., Dutta, P.S., *Methods for identifying critical transitions during cancer progression, (2025).* **To appear in "Cancer Systems Biology", Edited Book, Oxford University Press**

Participation in Conference/ Seminar / Workshop / Schools

2026 Participated in the **"Complexity72h Workshop 2026"**, held at Northeastern University London, Devon House campus, in London, United Kingdom, during 22–26 June 2026.

2023 Contributed Talk in **"NADCOM23: International Seminar and Workshop on Non-autonomous Dynamics in Complex Systems: Theory and Applications to Critical Transitions"** held at Max Planck Institute for the Physics of Complex Systems (MPIPKS) in Dresden during 09 - 27 October 2023.

2022 Poster presentation in **"Tipping Points in Complex Systems"** held at ICTS-TIFR, Bengaluru, India during 19-30 September 2022.

2022 Poster presentation in **"ESA & CSEE Joint Meeting Montréal, Québec, Canada"** by Ecological Society of America to be held on August 14 – 19, 2022.

2020 Poster presentation in **"ESA Annual Meeting 2020 Salt Lake City, Utah"** by Ecological Society of America to be held on August 3– 6, 2020.

2019 Poster presentation in in conference **"CNSD Conference"** held in IIT, Kanpur from December 11 - 15,2019.

2019 Poster presentation in **"Mathematical and Statistical explorations in disease modelling and public health (DMPH 2019)"** held at ICTS-TIFR, Bengaluru, India during 1-11 July 2019.

2018 Participated in **"IFCAM Summer School"** held in IISC Bangalore from 17th July to 31st July,2018 and supported by Indo-French Centre for the Promotion of Advanced Research (CEFIPRA), India.

References

1. **Prof. Marc Timme**,

   Chair of Network Dynamics, Institute of Theoretical Physics and Center for Advancing Electronics Dresden (cfaed)

   TUD Dresden University of Technology,

   01062 Dresden, Germany

   Homepage: https://cfaed.tu-dresden.de/cfnd-about

   E-Mail: marc.timme@tu-dresden.de

2. **Prof. Jordi Bascompte**,

   Department of Evolutionary Biology and Environmental Studies

   University of Zurich,

   Winterthurerstrasse 190, 8057 Zurich, Switzerland.

   Homepage: https://www.bascompte.net

   E-Mail: jordi.bascompte@uzh.ch

3. **Dr. Partha Sharathi Dutta**,

   Department of Mathematics,

   Indian Institute of Technology Ropar,

   Nangal Road, Rupnagar Punjab, India - 140 001.

   Homepage: https://mathbio.iitrpr.ac.in

   E-Mail: parthasharathi@iitrpr.ac.in